PROJECT ALPHA

24 MW Tier-III Data Centre - Chakan MIDC Phase II, Pune

480 V Low-Voltage Switchgear Package

Technical Specification, Single-Line Diagram & Equipment Schedule

Document No. PA-EL-SPC-480-001 Revision B

Rev	Date	Description	By	Chk
A	2026-11-02	Issued for review	RK	SD
B	2026-12-14	Issued for tender (RFQ)	RK	SD

Required on site (energisation window): 2027-07-30 | PO to be placed by: 2027-03-15

SECTION 26 23 00 - GENERAL REQUIREMENTS

1. Scope. Design, manufacture, type-test, factory acceptance test (FAT), supply and delivery to site of the 480 V low-voltage switchgear package for Data Hall Blocks A and B, including main lineups, bus-tie breakers, feeder breakers, automatic transfer switches, metering, surge protection and copper busway as listed in the Equipment Schedule (Sheet 4).

2. System. 480 V, 3-phase, 3-wire + PE, 50 Hz, TN-S. Short-circuit withstand 65 kA for 1 s. Main busbar 4000 A copper, tin-plated. Form 4b type 6 internal separation. IP42 indoor.

3. Standards. IEC 61439-1/-2 (type-tested assemblies), IEC 60947-2 (breakers), IEC 60947-6-1 (ATS), IEC 61643-11 (SPD), IEC 61439-6 (busway), IS/IEC 60947 where applicable. Arc-fault containment per IEC TR 61641, 65 kA 0.5 s.

4. Redundancy. 2N topology: Block A and Block B lineups each fed from a dedicated 3000 kVA transformer; normally-open tie breakers allow either transformer to carry both lineups.

5. Submittals. General arrangement, single-line, wiring and protection coordination study within 14 days of PO; Engineer review period 10 days. Manufacture shall not start before approval of shop drawings.

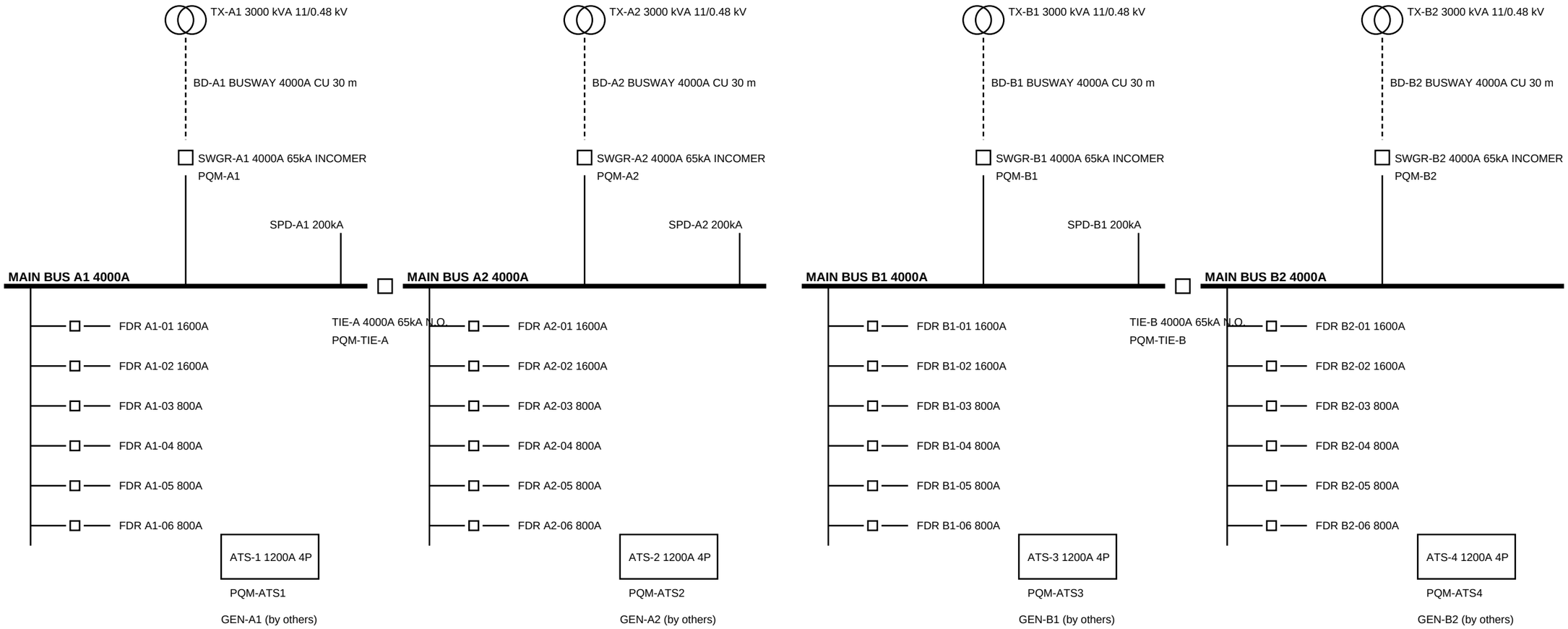
6. Testing. Routine tests per IEC 61439. Witnessed FAT (5 days) at the manufacturer's works with Owner and Engineer. Type-test certificates from an accredited lab (CPRI / KEMA / ASTA).

7. Delivery. Deliver DAP site, Chakan MIDC Phase II, Pune 410501, in shipping splits not exceeding 2.4 m, on or before 2027-07-30 (energisation window). Liquidated damages 0.5 % per week of delay, capped at 10 % of order value.

8. Warranty. 24 months from energisation or 30 months from delivery, whichever is earlier. Critical spares list and 5-year spares pricing to be submitted with the offer.

9. Commercial. Prices firm in INR ex-works, freight quoted separately, GST extra at 18 %. Quote validity minimum 14 days. Bidders shall state lead time from shop-drawing approval.

SINGLE-LINE DIAGRAM - 480 V SWITCHGEAR, DATA HALL BLOCKS A & B



NOTES: 1. Breaker symbols: square = draw-out ACB / MCCB. 2. Busway runs BD-xx are 4000 A copper, lengths indicative (verify on site).

3. Device ratings per Equipment Schedule, Sheet 4. Where drawing and schedule differ, the schedule governs and an RFI shall be raised.

EQUIPMENT SCHEDULE - 480 V SWITCHGEAR PACKAGE

Table 1. Quantities for tender. Unit 'ea' = each, 'm' = linear metre.

Item	Item Code	Description	Rating	Qty	Unit	Remarks
1	SWGR-MAIN-4000A	Main LV switchgear lineup, draw-out ACB incomer	4000 A, 65 kA	4	ea	SWGR-A1/A2/B1/B2
2	BKR-TIE-4000A	Bus-tie air circuit breaker, draw-out	4000 A, 65 kA	2	ea	TIE-A, TIE-B
3	BKR-FDR-1600A	Feeder ACB, electronic trip LSIG	1600 A, 65 kA	8	ea	UPS / chiller feeders
4	BKR-FDR-800A	Feeder MCCB, electronic trip LSI	800 A, 65 kA	16	ea	PDU / mech feeders
5	ATS-1200A-4P	Automatic transfer switch, 4-pole, open transition	1200 A	4	ea	ATS-1..4 gen backup
6	MTR-PQ-02S	Power quality meter, Modbus TCP	Class 0.2S	10	ea	Incomers, ties, ATS
7	SPD-T1-200KA	Surge protective device Type 1+2	200 kA	4	ea	One per main bus
8	BUSWAY-4000A-CU	Sandwich busway, copper, IP54	4000 A	120	m	TX to SWGR, 4 runs x 30 m

All breakers draw-out, motorised, with communication module. Spare feeder cubicles (2 per lineup) fully equipped are included in the lineup price.